

Lab Notebook, Summer 2026

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1. **Tuesday, May 19th:** Today, I spent most of my time documenting the Richards-Wolf integration code, and also building a pipeline to analyze the verification data when we take it tomorrow. There are a number of transformations required to properly compare the real and simulated data: translation in space, Poisson noise, and a potential linear (or nonlinear) scaling difference. The pipeline attempts to match these as best it can, producing a set of alignment parameters (with error bars). Then, using these parameters, we can generate many simulated images, and use them to construct a null distribution, from which we can assign a p-value to the real data. Before finishing work for the day, I also prepared a MATLAB script to run the experiment tomorrow, using the same aberrations that Diego used last summer. Finally, in anticipation of my machine learning work later in the summer, I spent an hour reading (and taking notes on) Joe Chang's *Stochastic Processes*, which I hope to complete before the end of June.
2. **Wednesday, May 20th:** From 9 AM to 1 PM, Shengqi and I worked on the 3P microscope trying to take data to check the Richards-Wolf integration code. Earlier, I had written code that would sequentially apply each of the six aberrations that Diego used last summer, and measure the PSF, a process that could be iterated as the z-level of the sample was adjusted. However, we ran into several serious issues. First, we noticed that the applied Zernike modes were not centered within the beam pupil, which we were able to quickly fix. Additionally, we had to figure out how to add a grating to the background of the applied SLM signal. Finally, we noticed that the aberrations were different in magnitude from the Python simulations. This was fixed by removing the "normalization" procedure, which was used previously to control the RMS of the aberration. In the afternoon, I finished making alterations to the code and testing it. Then, I proceeded to finish the Monte-Carlo pipeline I began yesterday, successfully running it to completion on simulated data. However, it was noted that the resulting p-statistic is extremely sensitive to differences in the experimental data and simulation, meaning that alterations may be necessary to apply it to real data. Finally, I spent an hour taking notes on *Stochastic Processes*.

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